- It only takes two parameters, that can also be derived by the domain (possibly through a preliminary scan to collect statistics on the dataset);
- DBSCAN can handle noise in the datasets by identifying outliers;
- It can find arbitrarily shaped clusters, and can partition non-linearly separable clusters (see figure 12.12, and compare it to figure 12.6 for k-means);
- By tuning the `minPoints` parameter, is possible to reduce the single-link effect;
- The algorithm is almost entirely deterministic and the order in which points are processed is mostly irrelevant: a different order can only change the assignment of points on the edge of clusters, when they are equally close to more than one cluster (as we have seen in figure 12.11).

So far for the good news; as you can imagine, every rose has its thorns, and DBSCAN has some shortcomings as well:

- DBSCAN is *almost* entirely deterministic, but not completely: for some applications it might be a problem that points at the border of two or more clusters are aleatorily assigned to one or another.
- DBSCAN also suffers from the curse of dimensionality: if the metric used is the Euclidean distance then, as we have seen in section 12.2.2, in high-dimensional spaces all neighbors of a point are at the same distance, and the distance function becomes basically useless. Luckily, other metrics can also be used with DBSCAN;
- If a dataset has areas with different densities, it becomes challenging, or sometimes impossible, to choose